Omar Ahmed

Firewall Evasion Lab

Tasks:[https://seedsecuritylabs.org/Labs_20.04/Files/Firewall_Evasion/Firewall_Evasion.pdf]

Task 0:

```
[11/28/24]seed@VM:~/.../Labsetup$ nslookup reddit.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   reddit.com
Address: 151.101.129.140
Name:   reddit.com
Address: 151.101.193.140
Name:   reddit.com
Address: 151.101.65.140
Name:   reddit.com
Address: 151.101.1.140
```

(4 ipv4 to block)

```
[11/28/24]seed@VM:~/.../Labsetup$ nslookup instagram.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   instagram.com
Address: 31.13.80.174
```

(1 ipv4 to block)

- Using nslookup command to find IP address of two websites: reddit.com and instagram.com

```
command: bash -c "
        ip route del default  &&
        ip route add default via 10.8.0.1 &&
        iptables -t nat -A POSTROUTING ! -d 10.8.0.0/24 -j MASQUERADE -o eth0 &&
        iptables -A FORWARD -i eth0 -p tcp -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEP
        iptables -A FORWARD -i eth0 -p tcp --dport 22 -j ACCEPT &&
        iptables -A FORWARD -i eth0 -p tcp -j DROP &&
        iptables -A FORWARD -i eth1 -d 93.184.216.0/24 -j DROP &&
        iptables -A FORWARD -i eth1 -d 151.101.129.140 -j DROP &&
        iptables -A FORWARD -i eth1 -d 151.101.193.140 -j DROP &&   reddit.com
        iptables -A FORWARD -i eth1 -d 151.101.65.140 -j DROP &&
        iptables -A FORWARD -i eth1 -d 151.101.1.140 -j DROP &&
        iptables -A FORWARD -i eth1 -d 31.13.80.174 -j DROP &&  ---> instagram.com
        iptables -A FORWARD -i eth1 -d 93.184.216.0/24 -j DROP &&
        /etc/init.d/openbsd-inetd start  &&
        tail -f /dev/null
    "
```

- Add the rules

```
[11/28/24]seed@VM:~/.../Labsetup$ dockps
a8cec30eb8f8  A1-10.8.0.5
b1b705e968e8  B2-192.168.20.6
08ff23338b62  B-192.168.20.99
4d0d1c512d29  A-10.8.0.99
01067bd89970  A2-10.8.0.6
be6c97f7d9f7  B1-192.168.20.5
de7b810ac1a2  router-firewall
[11/28/24]seed@VM:~/.../Labsetup$ docksh b1b
root@b1b705e968e8:/# ping instagram.com
PING instagram.com (31.13.80.174) 56(84) bytes of data.
^C
--- instagram.com ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3069ms

root@b1b705e968e8:/# ping reddit.com
PING reddit.com (151.101.65.140) 56(84) bytes of data.
^C
--- reddit.com ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1032ms

root@b1b705e968e8:/#
```

- Successfully blocked reddit.com and instagram.com from B2
    - Pings not successful

Task 1:

```
root@4d0d1c512d29:/# ssh -4NT -L 0.0.0.0:2223:192.168.20.5:23 root@192.168.20.
root@192.168.20.99's password:
```

- ssh command used on host A above
- Using port 2223 and IP 0.0.0.0 to listen on all interfaces

telnet into B1 from host A:

```
[11/28/24]seed@VM:~/.../Labsetup$ docksh 4d
root@4d0d1c512d29:/# telnet 127.0.0.1 2223
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
be6c97f7d9f7 login: root
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Nov 28 18:04:27 UTC 2024 from B-192.168.20.99.net-192.168.20.0 on pts
root@be6c97f7d9f7:~#
```

telnet into B1 from host A1:

```
root@a8cec30eb8f8:/# telnet 10.8.0.99 2223
Trying 10.8.0.99...
Connected to 10.8.0.99.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
be6c97f7d9f7 login: root
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Nov 28 18:05:28 UTC 2024 from B-192.168.20.99.net-192.168.20.0 on pt
root@be6c97f7d9f7:~#
```

telnet into B1 from host A2:

```
root@01067bd89970:/# telnet 10.8.0.99 2223
Trying 10.8.0.99...
Connected to 10.8.0.99.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
be6c97f7d9f7 login: root
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Thu Nov 28 18:07:18 UTC 2024 from B-192.168.20.99.net-192.168.20.0 on pt
root@be6c97f7d9f7:~#
```

How many TCP connections are involved in this entire process. You should run wireshark or tcpdump to capture the network traffic, and then point out all the involved TCP connections from the captured traffic.

```
a8cec30eb8f8  A1-10.8.0.5
b1b705e968e8  B2-192.168.20.6
08ff23338b62  B-192.168.20.99
4d0d1c512d29  A-10.8.0.99
01067bd89970  A2-10.8.0.6
be6c97f7d9f7  B1-192.168.20.5
de7b810ac1a2  router-firewall
```

tcpdump from host A:

```
18:18:44.350810 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.351183 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.351197 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:44.403592 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:44.403604 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.404222 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.404240 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:44.481527 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:44.481925 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.481954 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:44.483203 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:44.483259 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:48.158539 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
18:18:48.159055 IP 4d0d1c512d29.2223 > A2-10.8.0.6.net-10.8.0.0.59958:
18:18:48.159074 IP A2-10.8.0.6.net-10.8.0.0.59958 > 4d0d1c512d29.2223:
```

- 1 TCP connection from A1 or A2 (in this case it is A2) to A (4d0d1c512d29.2223)
- 1 SSH TCP connection from A to B

tcpdump from host B:

```
18:18:44.351127 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:44.403741 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:44.403769 IP B1-192.168.20.5.net-192.168.20.0.telnet > 08ff23338b62.552
18:18:44.404111 IP B1-192.168.20.5.net-192.168.20.0.telnet > 08ff23338b62.552
18:18:44.404117 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:44.481855 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:44.481829 IP B1-192.168.20.5.net-192.168.20.0.telnet > 08ff23338b62.552
18:18:44.481837 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:44.483123 IP B1-192.168.20.5.net-192.168.20.0.telnet > 08ff23338b62.552
18:18:44.483132 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
18:18:48.158683 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
 NAWS IS 0xc0 0 0x25 SE [|telnet]
18:18:48.158966 IP B1-192.168.20.5.net-192.168.20.0.telnet > 08ff23338b62.552
18:18:48.158977 IP 08ff23338b62.55208 > B1-192.168.20.5.net-192.168.20.0.teln
```

- 1 TCP connection from B (08bff23338b62.55208) to B1
- 3 connections total involved in the entire process.

Why can this tunnel successfully help users evade the firewall rule specified in the lab setup?

- The firewall blocks outside machines from internal ones, other than the SSH server.
- By port forwarding, traffic is hidden as an SSH connection

Task 2:

```
root@47b6d54a0295:/# nslookup facebook.com
Server:         127.0.0.11
Address:        127.0.0.11#53

Non-authoritative answer:
Name:   facebook.com
Address: 31.13.80.36
Name:   facebook.com
Address: 2a03:2880:f10e:83:face:b00c:0:25de
```

2.1:

Block instagram and facebook:

```
root@47b6d54a0295:/# nslookup www.instagram.com
Server:         127.0.0.11
Address:        127.0.0.11#53

Non-authoritative answer:
www.instagram.com       canonical name = z-p42-instagram.c10r.instagram.com.
Name:   z-p42-instagram.c10r.instagram.com
Address: 31.13.80.174
Name:   z-p42-instagram.c10r.instagram.com
Address: 2a03:2880:f20e:e5:face:b00c:0:4420
```

```
root@3b4e939c12b1:/# iptables -A FORWARD -i eth1 -d 31.13.80.36 -j DROI
root@3b4e939c12b1:/# iptables -A FORWARD -i eth1 -d 31.13.80.174 -j DR(
```

Ping from inside container B:

```
root@29a675e098fa:/# ping 93.184.216.34
PING 93.184.216.34 (93.184.216.34) 56(84) bytes of data.
^C
--- 93.184.216.34 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2056ms
```

```
root@29a675e098fa:/# ping www.instagram.com
PING z-p42-instagram.c10r.instagram.com (31.13.80.174) 56(84) bytes of data.
^C
--- z-p42-instagram.c10r.instagram.com ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2040ms
```

```
^Croot@29a675e098fa:/# ssh -4NT -D 0.0.0.0:1080 root@10.8.0.99
root@10.8.0.99's password:
```

```
root@29a675e098fa:/# curl -v --proxy socks5h://192.168.20.99:1080 www.instagram.com
*   Trying 192.168.20.99:1080...
* TCP_NODELAY set
* SOCKS5 communication to www.instagram.com:80
* SOCKS5 connect to www.instagram.com:80 (remotely resolved)
* SOCKS5 request granted.
* Connected to 192.168.20.99 (192.168.20.99) port 1080 (#0)
> GET / HTTP/1.1
> Host: www.instagram.com
> User-Agent: curl/7.68.0
> Accept: */*
>
* Mark bundle as not supporting multiuse
< HTTP/1.1 301 Moved Permanently
< Location: https://www.instagram.com/
< Content-Type: text/plain
< Server: proxygen-bolt
< Date: Mon, 02 Dec 2024 04:48:01 GMT
< Connection: keep-alive
< Content-Length: 0
<
* Connection #0 to host 192.168.20.99 left intact
root@29a675e098fa:/#
```

(1) Which computer establishes the actual connection with the intended web server?

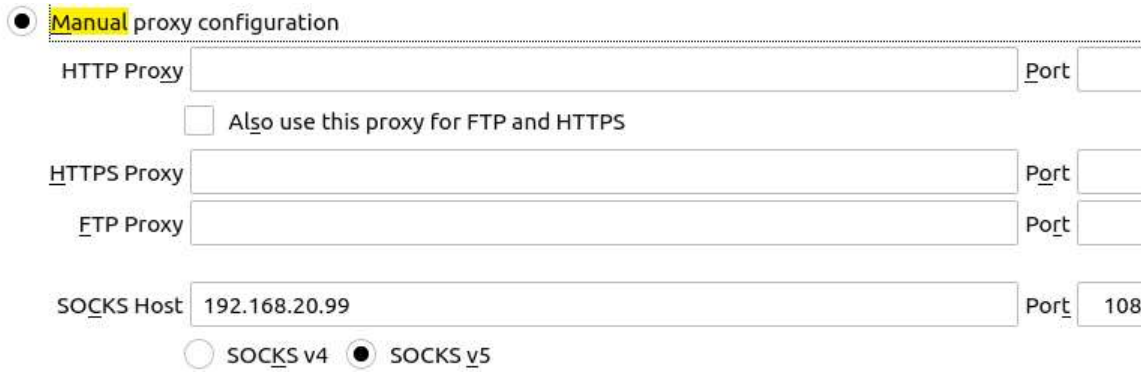
- Host A establishes the connection with the external web server because the SSH tunnel routes traffic from B to A.

(2) How does this computer know which server it should connect to?

- Host A knows which server to connect to based on the destination information provided by the SOCKS proxy on Host B.

2.2:

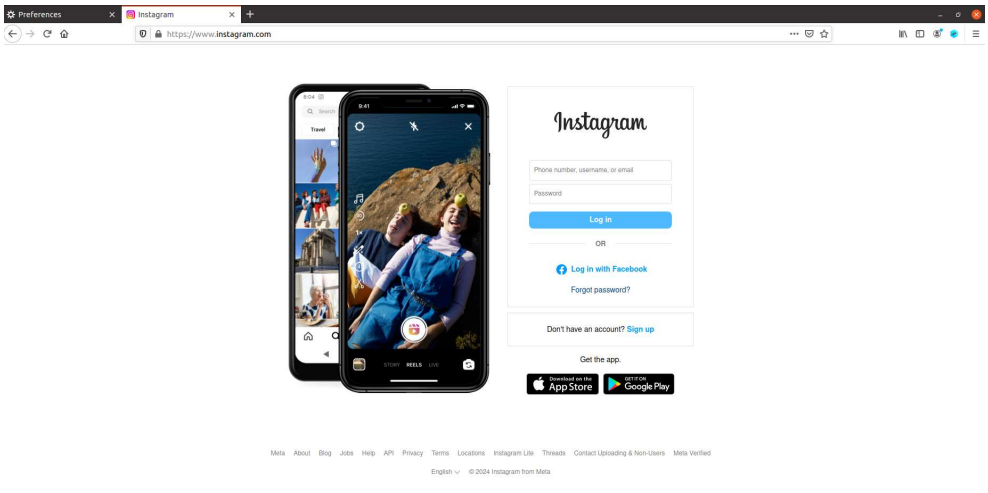
Firefox proxy config:



Now we start monitoring traffic on the router:

```
root@3b4e939c12b1:/# tcpdump -i eth0 host www.instagram.com
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

```
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
05:03:05.063250 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [P.], seq 248308850:248309246, ack 80126
 win 18395, options [nop,nop,TS val 2074171444 ecr 3796173815], length 396
05:03:05.063264 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [.], ack 396, win 623, options [nop,nop,T
l 3796180678 ecr 2074171444], length 0
05:03:05.064948 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [P.], seq 396:1632, ack 1, win 18395, opt
 [nop,nop,TS val 2074171446 ecr 3796180678], length 1236
05:03:05.064955 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [.], ack 1632, win 620, options [nop,nop,
al 3796180680 ecr 2074171446], length 0
05:03:05.065356 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [P.], seq 1632:2156, ack 1, win 18395, op
s [nop,nop,TS val 2074171447 ecr 3796180680], length 524
05:03:05.065363 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [.], ack 2156, win 623, options [nop,nop,
al 3796180681 ecr 2074171447], length 0
05:03:05.085363 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [P.], seq 2156:2352, ack 1, win 18395, op
s [nop,nop,TS val 2074171467 ecr 3796180681], length 196
05:03:05.085378 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [.], ack 2352, win 624, options [nop,nop,
al 3796180701 ecr 2074171467], length 0
05:03:05.193150 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [P.], seq 1:85, ack 2352, win 624, optior
op,nop,TS val 3796180808 ecr 2074171467], length 84
05:03:05.193173 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [.], ack 85, win 18395, options [nop,nop,
al 2074171574 ecr 3796180808], length 0
05:03:05.220790 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [P.], seq 85:2993, ack 2352, win 624, opt
 [nop,nop,TS val 3796180836 ecr 2074171574], length 2908
05:03:05.220811 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [.], ack 2993, win 18389, options [nop,no
 val 2074171602 ecr 3796180836], length 0
05:03:05.239981 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [P.], seq 2993:10233, ack 2352, win 624,
ons [nop,nop,TS val 3796180855 ecr 2074171602], length 7240
05:03:05.240002 IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh: Flags [.], ack 10233, win 18372, options [nop,n
S val 2074171621 ecr 3796180855], length 0
05:03:05.240007 IP A-10.8.0.99.net-10.8.0.0.ssh > B-192.168.20.99.net-192.168.20.0.56284: Flags [P.], seq 10233:17473, ack 2352, win 624,
ions [nop,nop,TS val 3796180855 ecr 2074171602], length 7240
```

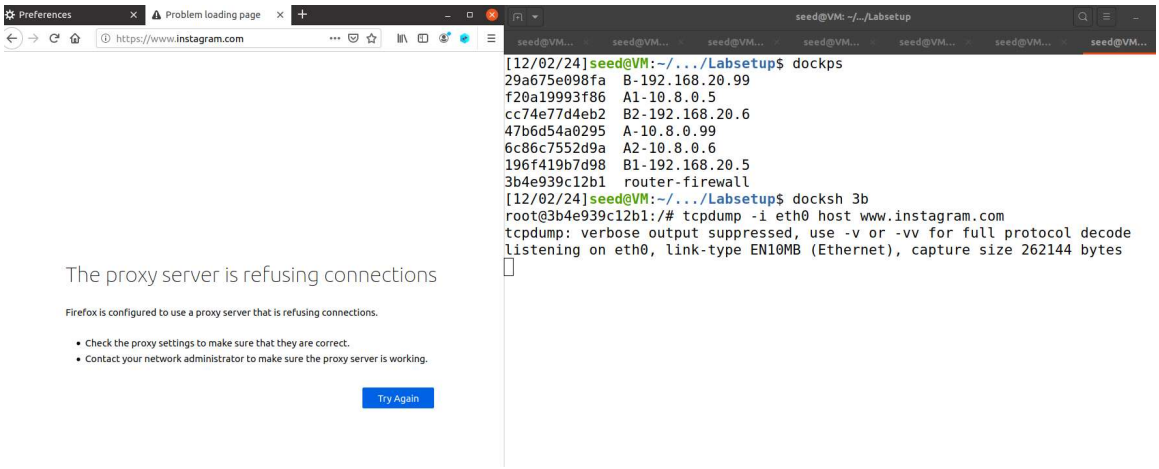


Tcpdump outputs: “IP B-192.168.20.99.net-192.168.20.0.56284 > A-10.8.0.99.net-10.8.0.0.ssh”

- This indicates that traffic is being sent from host B to host A when accessing www.instagram.com
- This means that a tunnel has been established and is working properly

Now lets break the SSH tunnel and test again:

```
^Croot@29a675e098fa:/# ssh -4NT -D 0.0.0.0:1080 root@10.8.0.99
root@10.8.0.99's password:
^Croot@29a675e098fa:/#
```



As you can see, Instagram is blocked once again.

2.3:

socks_client.py:

```python
#!/bin/env python3

import socks

s = socks.socksocket()
s.set_proxy(socks.SOCKS5, "192.168.20.99", 1080)

hostname = "www.instagram.com"
s.connect((hostname, 80))

request = b"GET / HTTP/1.0\r\nHost: " + hostname.encode('utf-8') + b"\r\n\r\n"
s.sendall(request)

response = s.recv(2048)
while response:
    print(response.split(b"\r\n"))
    response = s.recv(2048)

s.close()

```

Restart ssh:

```
root@29a675e098fa:/# ssh -4NT -D 0.0.0.0:1080 root@10.8.0.99
root@10.8.0.99's password:
```

Install pysocks (for import socks) and run the script:

```
[12/02/24]seed@VM:~/.../Labsetup$ pip3 install PySocks
Collecting PySocks
  Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)
Installing collected packages: PySocks
Successfully installed PySocks-1.7.1
[12/02/24]seed@VM:~/.../Labsetup$ python3 socks_client.py
[b'HTTP/1.1 301 Moved Permanently', b'Location: https://www.instagram.com/', b'Content-Type: text/plain', b'Server: proxygen-bolt', b'Dat
on, 02 Dec 2024 05:27:11 GMT', b'Connection: close', b'Content-Length: 0', b'', b'']
```

- We get redirected to https likely because instagram.com automatically redirects to https when trying to access http.
- This indicates that the script and port-forwarding work

Task 3:

```
[12/02/24]seed@VM:~/.../Labsetup$ dockps
b17d6073e564  A1-10.8.0.5
9adeaaedd783  B-192.168.20.99
c1a13ae93fd0  router-firewall
4326b5168f5f  A-10.8.0.99
c3476300057f  B2-192.168.20.6
bc0508bcf9b9  A2-10.8.0.6
732f9704e2e1  B1-192.168.20.5
[12/02/24]seed@VM:~/.../Labsetup$ docksh 432
root@4326b5168f5f:/# ssh -w 0:0 root@192.168.20.99 -o "PermitLocalCommand=yes" -o "LocalCommand= ip addr add
68.53.88/24 dev tun0 && ip link set tun0 up" -o "RemoteCommand=ip addr add 192.168.53.99/24 dev tun0 && ip li
t tun0 up"
root@192.168.20.99's password:
```

- Run on Host A to establish the VPN tunnel with Host B
- This command sets up the TUN interfaces and assigns IPs to enable communication between the hosts
- Allows communication between the external and internal networks by bypassing the ingress firewall restrictions

```
[12/02/24]seed@VM:~/.../Labsetup$ dockps
b17d6073e564  A1-10.8.0.5
9adeaaedd783  B-192.168.20.99
c1a13ae93fd0  router-firewall
4326b5168f5f  A-10.8.0.99
c3476300057f  B2-192.168.20.6
bc0508bcf9b9  A2-10.8.0.6
732f9704e2e1  B1-192.168.20.5
[12/02/24]seed@VM:~/.../Labsetup$ docksh 732
root@732f9704e2e1:/# telnet 10.8.0.99
Trying 10.8.0.99...
Connected to 10.8.0.99.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
4326b5168f5f login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
seed@4326b5168f5f:~$ exit
logout
Connection closed by foreign host.
root@732f9704e2e1:/# exit
exit
[12/02/24]seed@VM:~/.../Labsetup$ docksh c34
root@c3476300057f:/# telnet 10.8.0.5
Trying 10.8.0.5...
Connected to 10.8.0.5.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
b17d6073e564 login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

seed@b17d6073e564:~$
```

- Telnet command used inside Host B to connect to Host A on the external network
- Connection was successfully established
- Normally the Ingress firewall would block connections from Host B (internal) to Host A (external) but this was able to bypass the firewall and establish communication

```
[12/02/24]seed@VM:~/.../Labsetup$ dockps
b17d6073e564  A1-10.8.0.5
9adeaaedd783  B-192.168.20.99
c1a13ae93fd0  router-firewall
4326b5168f5f  A-10.8.0.99
c3476300057f  B2-192.168.20.6
bc0508bcf9b9  A2-10.8.0.6
732f9704e2e1  B1-192.168.20.5
[12/02/24]seed@VM:~/.../Labsetup$ docksh 9ad
root@9adeaaedd783:/# ssh -w 0:0 root@10.8.0.99 -o "PermitLocalCommand=yes" -o "LocalCommand= ip addr add 192.1
3.99/24 dev tun0 && ip link set tun0 up" -o "RemoteCommand=ip addr add 192.168.53.88/24 dev tun0 && ip link se
n0 up"
root@10.8.0.99's password:
```

- Created a VPN Tunnel from Host B to Host A
- Establishes VPN Tunnel between Host B (Internal) and host A (external), enabling Host B to route traffic through Host A to bypass the egress firewall

```
[12/02/24]seed@VM:~/.../Labsetup$ dockps
b17d6073e564  A1-10.8.0.5
9adeaaedd783  B-192.168.20.99
c1a13ae93fd0  router-firewall
4326b5168f5f  A-10.8.0.99
c3476300057f  B2-192.168.20.6
bc0508bcf9b9  A2-10.8.0.6
732f9704e2e1  B1-192.168.20.5
[12/02/24]seed@VM:~/.../Labsetup$ docksh c34
root@c3476300057f:/# telnet 10.8.0.6
Trying 10.8.0.6...
Connected to 10.8.0.6.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
bc0508bcf9b9 login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

seed@bc0508bcf9b9:~$
```

- Validates that traffic from internal hosts can traverse through the VPN tunnel to connect with other hosts, bypassing the egress firewall that would otherwise block it.

```
root@96e8dcf1a24a:/# iptables -t nat -A POSTROUTING -j MASQUERADE -o eth0
root@96e8dcf1a24a:/# curl www.example.com
<!doctype html>
<html>
<head>
    <title>Example Domain</title>

    <meta charset="utf-8" />
    <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <style type="text/css">
    body {
        background-color: #f0f0f2;
        margin: 0;
        padding: 0;
        font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica
e", Helvetica, Arial, sans-serif;

    }
    div {
        width: 600px;
        margin: 5em auto;
        padding: 2em;
        background-color: #fdfdff;
        border-radius: 0.5em;
        box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
        color: #38488f;
        text-decoration: none;
    }
    @media (max-width: 700px) {
        div {
            margin: 0 auto;
            width: auto;
        }
    }
    </style>
</head>

<body>
<div>
    <h1>Example Domain</h1>
    <p>This domain is for use in illustrative examples in documents. You may use this
    domain in literature without prior coordination or asking for permission.</p>
    <p><a href="https://www.iana.org/domains/example">More information...</a></p>
</div>
</body>
</html>
root@96e8dcf1a24a:/#
```

- NAT is configured on A to modify the source IP address of packets originating from B before they are sent to external websites
- Ensures the egress firewall at A's side does not block traffic, as it sees packets originating from a legitimate source (A).
- Allows B to receive the response without any interference from firewalls

Task 4:

- Scope
  - SOCKS5 proxy works at the application level, forwarding specific app traffic
  - VPN operates at the network level, encrypting and routing traffic through the tunnel
- Encryption
  - SOCKS5 does not encrypt traffic (vulnerable)
  - VPN provides encryption for enhanced security (more secure)
- Setup/Performance
  - SOCKS5 is simple and faster
  - VPN requires more setup and has more overhead because of encryption
- Use Cases
  - SOCKS5 better for lightweight, app-specific tasks like bypassing firewalls for a single program
  - VPN is better for more intense security, bypassing firewalls at the network level, and accessing private networks securely

Overall, SOCKS5 is faster and simpler but lacks security, whale VPN is more secure but at the cost of complexity and performance